

Midway Report: Citation-Grounded Scientific QA with Local Retrieval-Augmented Generation

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Abstract

We study hallucination reduction in scientific question answering with a fully local retrieval-augmented generation (RAG) pipeline. Our setting uses QASPER as the benchmark, Phi-3.5-mini-instruct as the main generator, and shared Boston University SCC infrastructure for reproducible experiments. By the midway point, we have completed the data pipeline, chunking strategies, retrieval stack, and the proposal’s core generation ablations A–E, along with three additional baselines: parametric-only generation, BM25-only retrieval, and random retrieval. On the original 100-question validation sample, the strongest core condition is section-aware chunking with hybrid retrieval and baseline prompting (Condition C), which reaches token F1 0.3947, BERTScore F1 0.8946, hallucination rate 0.39, and supported-answer rate 0.79. We subsequently extended the strongest settings to full validation and added an NLI-based citation check: on 1005 validation questions, the best overall Phi baseline (Condition D) reaches token F1 0.3238, while citation forcing (Condition E) reaches token F1 0.3215 with citation precision 0.3230, citation rate 0.9274, and NLI citation precision 0.1416. Citation forcing improves explicit attribution behavior, but it still reduces grounding relative to the strongest baseline setting.

1 Introduction

Our project asks how much hallucination in a small, locally hosted language model is caused by retrieval quality versus generation behavior. We focus on scientific question answering because scientific answers are high-stakes, evidence-dependent, and naturally compatible with citation-grounded evaluation. We use QASPER (Dasigi et al., 2021), which provides research-paper questions, answers, and gold evidence spans, allowing

us to evaluate both retrieval and final answers.

This project draws most directly on course material from the transformer and LLM lectures (attention, pretraining, prompting, and local adaptation), the question answering and information retrieval lecture, and the evaluation lectures on metrics and best practices. Several engineering components in our system were not covered directly in lecture, including FAISS indexing, reciprocal-rank fusion, cross-encoder reranking, and citation-grounded RAG evaluation. We therefore learned those pieces from project readings, model documentation, and system-building practice. We explicitly note these gaps because they shape the remaining work before the final report.

2 Description of Progress

2.1 Methods

We implemented a fully local RAG pipeline around QASPER. The data layer normalizes raw papers, questions, answers, and evidence into a shared internal schema. We then build deterministic processed splits in a shared SCC course directory so the entire team uses identical artifacts.

For chunking, we implemented the three strategies proposed in our project plan: fixed-size chunks (256 tokens, 32 overlap), sliding-window chunks (512 tokens, 128 overlap), and section-aware chunks aligned to paper structure. For retrieval, we implemented BM25, dense retrieval with BGE-small (Xiao et al., 2023), hybrid reciprocal-rank fusion, and hybrid retrieval with cross-encoder reranking. For generation, we use microsoft/Phi-3.5-mini-instruct (Microsoft, 2024) with two prompt styles: baseline answering and citation forcing, where answers must explicitly cite retrieved passages. The evaluation pipeline measures token F1, BERTScore, citation

Method	Recall@5	MRR@10	Evidence hit
BM25	0.5769	0.4110	0.7152
Dense	0.5886	0.4820	0.7238
Hybrid	0.6313	0.5097	0.7778
Hybrid + rerank	0.6410	0.5027	0.7702

Table 1: Validation retrieval results on section-aware chunks.

precision/rate, NLI-based citation precision, hallucination rate, supported-answer rate, Recall@5, MRR@10, and evidence hit rate.

2.2 Experiments

We completed the retrieval experiments on validation section chunks and then ran the proposal’s core ablation conditions:

- A: fixed chunking + dense retrieval + baseline prompting
- B: fixed chunking + hybrid retrieval + baseline prompting
- C: section-aware chunking + hybrid retrieval + baseline prompting
- D: section-aware chunking + hybrid retrieval + reranking + baseline prompting
- E: section-aware chunking + hybrid retrieval + reranking + citation forcing

We also added the proposal baselines: parametric-only generation (no retrieval), BM25-only retrieval, and random retrieval over five passages. All generation conditions above were first run on a 100-question validation sample to keep experiments comparable on SCC. After stabilizing the pipeline, we extended A–E, parametric-only, and BM25-only to the full 1005-question validation split and added an NLI-based citation scorer for citation-forcing runs. In addition, we ran exploratory model comparisons with Qwen2.5-7B and Qwen3-4B, but those are side experiments rather than the main midway result story.

2.3 Results

Table 1 summarizes the retrieval-only experiments. Hybrid retrieval gives the best overall validation balance, while hybrid plus reranking improves Recall@5 slightly but does not dominate every metric.

Table 2 shows the main A–E generation results. The strongest condition so far is C: section-aware chunking with hybrid retrieval and baseline prompting. Moving from fixed hybrid retrieval (B) to section-aware hybrid retrieval (C) gives the largest gain in the main table. Reranking (D) does not beat C on this 100-question sample, although it does slightly improve supported-answer rate. Citation forcing (E) produces citations very reliably, but still lowers answer quality and grounding relative to the strongest non-citation setting.

Table 3 gives the three extra baselines from the proposal. Parametric-only generation is much worse than any retrieval-based condition, which supports the central claim that retrieval materially reduces hallucination in this task. BM25-only performs better than random retrieval and nearly matches the weaker dense/hybrid ablations on this sample, while random retrieval remains clearly worse on grounding.

One practical lesson from these experiments is that token F1 alone understates answer quality. The stronger systems in our study have much higher BERTScore than token F1, showing that many predictions are semantically reasonable paraphrases even when they do not exactly match QASPER’s canonical wording. However, BERTScore does not replace grounding metrics, so we still report hallucination and supported-answer rate separately.

Table 4 summarizes the later full-validation follow-up on the strongest Phi settings. The best full-validation baseline is now D rather than C, but the overall story is unchanged: retrieval plus reranking helps the baseline system, while citation forcing still trades answer support for explicit attribution. The NLI-based citation result is especially important because it shows that many cited claims are still not actually entailed by the cited evidence. This confirms the error-analysis finding that explicit citation formatting alone is not enough to ensure grounded attribution.

3 Project Plan

Before the final report, we plan to do four things. First, we will complete the remaining proposal evaluation pieces that are not yet finished at report quality: a local-judge faithfulness metric such as RAGAS (Es et al., 2023), a small manual error analysis, and a lightweight human evaluation pass. Second, we will finish consolidating the full-

Cond.	Retrieval	Chunking	Prompt	Token F1	BERTScore F1	Halluc.
A	Dense	Fixed	Baseline	0.3636	0.8918	0.50
B	Hybrid	Fixed	Baseline	0.3365	0.8834	0.52
C	Hybrid	Section	Baseline	0.3947	0.8946	0.39
D	Hybrid + rerank	Section	Baseline	0.3697	0.8910	0.43
E	Hybrid + rerank	Section	Citation	0.3569	0.8871	0.54

Table 2: Core proposal ablations on a 100-question validation sample. Condition E is the citation-forcing setting, while Condition C is the strongest non-citation condition in this table.

Baseline	Token F1	BERTScore	Halluc.	Supported
Parametric-only	0.1600	0.8397	0.95	0.05
BM25-only	0.3426	0.8896	0.49	0.70
Random retrieval	0.3257	0.8856	0.68	0.46

Table 3: Additional baselines on the same 100-question validation sample.

Setting	Questions	Token F1	Cite Prec.	Cite Rate	NLI Cite
D full val.	1005	0.3238	—	—	—
E full val.	1005	0.3215	0.3230	0.9274	0.1416

Table 4: Full-validation follow-up on the strongest Phi settings. Citation-only columns are not applicable to D because baseline prompting does not produce citations.

validation A–E table and supporting baselines into the final writeup. Third, we will decide whether any additional citation-tuning experiment is still justified now that the NLI citation check is available. Fourth, we will update the writeup to clearly separate our core Phi-based ablation story from the exploratory Qwen side experiments.

We also want to be explicit about what is not yet covered relative to the proposal and course scope. We have not yet run the proposal’s stretch-generator conditions with Llama-3-8B or Mistral-7B. We also have not yet completed human evaluation or a local-judge faithfulness metric beyond the NLI-based attribution scorer. These missing pieces are not due to a conceptual shift in the project; they are deferred because the mid-way milestone was spent getting the local SCC pipeline, retrieval stack, full-validation Phi experiments, and attribution checks into a stable, reproducible state.

4 Conclusion

By the midway point, the core system is working end-to-end and the main proposal ablation table is substantially complete. The clearest result so far is that chunking and retrieval choices matter a great deal: section-aware chunking with hybrid retrieval gives the best initial sample results, while

the later full-validation follow-up favors hybrid retrieval with reranking under baseline prompting. Citation forcing is promising for attribution behavior, but the new NLI result shows that frequent citation formatting still does not guarantee grounded attribution. The remaining work is therefore focused less on infrastructure and more on tightening evaluation and improving grounded generation quality.

References

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